

TYLER ROBERTS

Senior Software Engineer – C# (.NET) Azure/AWS

tylerrobertsdev@outlook.com | linkedin.com/in/tyler-roberts-1454473b4 | Florida, United States | +1 (321) 376-7095

SUMMARY

Senior Software Engineer with ~10 years delivering **enterprise SaaS and fintech-grade backend platforms** using **.NET Core (3.1–8), C#, and ASP.NET Core APIs**, specializing in **distributed microservices, event-driven systems, and cloud-native architecture on AWS/Azure**, with a strong focus on **performance optimization, security (OAuth2/JWT), and observability**, consistently improving system reliability by 40%+ through **scalable API design, resilient messaging patterns, and production-grade CI/CD pipelines**.

SKILLS

- **Backend & APIs** — C#, .NET Core (3.1–8), ASP.NET Core Web API, Entity Framework Core, REST APIs, microservices, GraphQL, background processing, WebSockets, API versioning, idempotency, rate limiting
- **Data, Messaging & Search** — PostgreSQL, SQL Server, MySQL, Redis, MongoDB, DynamoDB, Kafka, RabbitMQ, AWS SQS/SNS, Elasticsearch, OpenSearch
- **Cloud & DevOps** — AWS (EKS, ECS, Lambda, API Gateway, S3, RDS), Azure (App Services, Functions, Service Bus), Docker, Kubernetes, Terraform, CI/CD pipelines, GitHub Actions
- **Observability, Quality & Security** — OpenTelemetry, Datadog, Prometheus, Grafana, Serilog, NUnit, xUnit, OAuth2, OpenID Connect, JWT, RBAC

EXPERIENCE

Senior Software Engineer

Web Venturez LLC

February 2020 - October 2025

- Architected **high-throughput microservices platform** using **.NET Core (5–8)** and **ASP.NET Core Web API**, enabling scalable multi-tenant SaaS operations while improving request throughput by over **45%** through optimized middleware and async pipelines.
 - Designed and implemented **event-driven architecture** using **Kafka and AWS SQS/SNS**, resolving prior message duplication bugs by introducing **idempotency keys and deduplication strategies**, reducing failure retries by **60%**.
 - Migrated legacy **.NET Framework monolith** to **.NET Core (3.1–6)** services, solving memory leak issues caused by improper DI lifecycles and improving system stability under load.
 - Built **secure authentication systems** using **OAuth2, OpenID Connect, and JWT**, addressing token replay vulnerabilities by implementing strict expiration and refresh token rotation.
 - Optimized **Entity Framework Core queries** and indexing strategies across **PostgreSQL and SQL Server**, reducing slow query latency by **35%** in production workloads.
 - Implemented **Redis caching layers** and distributed locking mechanisms, eliminating race condition bugs in concurrent order processing workflows.
 - Led implementation of **OpenTelemetry tracing with Datadog dashboards**, improving incident detection time and reducing MTTR by over **50%**.
 - Developed **background job processing pipelines** using worker services and queue consumers, fixing retry storm issues with exponential backoff and circuit breaker patterns.
 - Containerized services with **Docker and Kubernetes (EKS)**, resolving deployment inconsistencies and ensuring reproducible environments across staging and production.
 - Automated CI/CD pipelines using **GitHub Actions**, reducing deployment time by **70%** and eliminating manual release errors.
 - Introduced **rate limiting and API throttling**, preventing abuse scenarios and stabilizing high-traffic endpoints under peak load.
 - Collaborated across teams to define **API contracts and schema validation**, reducing integration bugs between frontend and backend systems.
-

Software Engineer

Webiz Digital

July 2017 - January 2020

- Developed **RESTful APIs using ASP.NET Core (2.1–3.1)** and integrated with React-based frontends, ensuring consistent data contracts and scalable service boundaries.
- Resolved critical production bug related to **thread blocking in synchronous database calls**, refactoring to async patterns which improved response times by **40%**.
- Implemented **JWT-based authentication and RBAC authorization**, securing endpoints and preventing unauthorized access across multi-role systems.
- Designed relational schemas and optimized queries in **MySQL and PostgreSQL**, eliminating N+1 query issues and improving reporting performance.
- Integrated **Elasticsearch** for advanced filtering and search capabilities, enabling near real-time data retrieval for analytics dashboards.
- Built **background job workers** for file processing and batch operations, fixing failure loops by implementing retry queues and dead-letter handling.
- Introduced **Docker-based local development environments**, reducing onboarding time for new developers and eliminating environment mismatch issues.
- Worked on **Azure App Services and Azure Functions**, enabling serverless workflows and improving cost efficiency for scheduled jobs.
- Enhanced API reliability by implementing **global exception handling and structured logging using Serilog**, improving debugging visibility.
- Participated in migration of monolithic modules into **modular service-based architecture**, improving maintainability and deployment flexibility.
- Collaborated in agile teams delivering features with **unit testing (xUnit)** and CI integration, improving code quality and reducing regression issues.

Software Developer

Centric Tech Inc.

October 2013 - June 2017

- Built backend systems using **C# and ASP.NET MVC / early ASP.NET Core**, supporting internal enterprise tools and customer-facing web applications.
- Developed **CRUD-based APIs and SQL Server stored procedures**, resolving data inconsistency bugs through transactional integrity improvements.
- Maintained legacy systems and performed incremental migration toward **.NET Core (1.0–2.0)**, ensuring backward compatibility during transition phases.
- Debugged production issues related to **thread contention and memory usage**, optimizing application performance through profiling and refactoring.
- Implemented **basic authentication and session-based security**, later upgraded to token-based approaches to improve scalability.
- Designed database schemas and indexing strategies, improving query execution time for reporting modules.
- Built internal tools for **data import/export and batch processing**, resolving file parsing errors and improving reliability of ETL processes.
- Introduced logging mechanisms for better issue tracking, reducing debugging time during production incidents.
- Collaborated with QA teams to identify and fix edge-case bugs, improving system stability across releases.
- Supported deployment pipelines using **manual and early CI tools**, gradually improving release consistency and reducing downtime.

EDUCATION

Bachelor's degree in Computer Science

Florida State University

2009 - 2013

- Completed coursework emphasizing **software engineering fundamentals, data structures, and database-backed application design**, providing a strong foundation for owning **production backend systems** end-to-end.